AIDCF-018-ENV Environmental and Resource Performance Domain Guide

Document control

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Revision history

Version	Date	Summary
1.0	2026-10-10	Initial issue: domain guide for the 24 ENV controls, gate linkage, parameters, evidence and testing-safety expectations.

1 Purpose and scope

This guide explains how the ENV domain of AIDCF-007A is applied in an AIDCF assessment. The ENV domain has two faces, and both matter for AI capability:

- **Environmental conditions as a dependency of capability**: the temperature, humidity, contamination, vibration and acoustic conditions in which AI hardware must operate, and the climate and resource conditions (heat, drought, flood, water and grid availability) that the facility must withstand over its life.
- **Environmental performance as a claim about capability**: the energy, water and carbon performance (PUE, WUE, CUE, ERF, REF), the handling of hazardous materials (batteries, refrigerants, coolants, fuel, suppressants) and waste, and the traceability of reported metrics to regulators, customers and investors.

In scope: environmental operating envelopes (ENV-01); energy performance metrics and the resilience-efficiency trade-off (ENV-02); water source, quality, use and discharge (ENV-03); carbon accounting boundaries, energy attribute evidence, embodied data and reporting traceability (ENV-04); battery materials, refrigerants and coolants, fuel and suppressants, hardware and media waste (ENV-05); and climate adaptation and long-term resource constraints (ENV-06).

Out of scope, with interfaces: the design and capacity of cooling plant (THM), fire protection (PHY), battery systems in service (POW), media sanitisation methods (STO), and supplier sustainability due diligence (SUP).

This guide is informative except where a requirement carries a DG-ENV-REQ identifier.

2 Why this domain matters for AI capability

Tighter coupling between environment and throughput. Accelerators respond to inlet air or coolant temperature by throttling clock frequency before they fail. A hall that is "within allowable" may still be delivering less training throughput than the Service Outcome assumes. Measuring temperatures where equipment experiences them (rack inlet; technology cooling system supply) is therefore an outcome control, not housekeeping (MCF-ENV-001 (Temperature Control)).

Condensation risk with liquid cooling. Cold plates and manifolds run at coolant temperatures that may approach or fall below room dew point, especially with lower-temperature facility water or during humid weather. The margin between coolant supply temperature and dew point is a new and critical environmental parameter (MCF-ENV-002 (Humidity Control)).

Acoustic and vibration effects. Dense halls with high-speed fans can exceed occupational noise action values; inert-gas suppression discharge can produce sound pressure high enough to disrupt hard disk drives in storage arrays; vibration from rotating plant can degrade storage throughput (MCF-ENV-004 (Vibration and Acoustic Conditions)).

Scale of resource demand. AI campuses can demand hundreds of megawatts and, with evaporative heat rejection, large volumes of water. Grid capacity, water entitlements and environmental permits become long-term constraints on capability growth (MCF-ENV-024 (Long-Term Resource Constraints)) and on continuity during drought or heatwave (MCF-ENV-011 (Water Scarcity Response), MCF-ENV-021 (Heatwave Readiness)).

Efficiency pressure can erode resilience. UPS eco-mode, raised supply temperatures, reduced running redundant units and aggressive economiser hours each improve PUE and can each reduce ride-through or thermal margin. AIDCF requires an explicit resilience impact assessment for each (MCF-ENV-008 (Resilience-Efficiency Trade-Off)).

Claims under scrutiny. Regulators (for example under the EU Energy Efficiency Directive data-centre reporting scheme), customers and investors increasingly rely on reported PUE, WUE, renewable share and emissions. Metrics must be measured per the ISO/IEC 30134 family and traceable to meters (MCF-ENV-005, MCF-ENV-010, MCF-ENV-016).

Hazardous materials at scale. More lithium-ion, more refrigerant charge (with a shift to lower-GWP but mildly flammable A2L refrigerants), more glycol and, for immersion, dielectric fluids: each brings safety and environmental obligations (MCF-ENV-017 (Battery Materials), MCF-ENV-018 (Refrigerants and Coolants)).

3 Reference patterns and variants

Pattern / variant	ENV characteristics	ENV questions that dominate
Air-cooled with evaporative heat rejection	High water use; good energy efficiency in dry climates	WUE; Legionella; water scarcity response
Air-cooled with economisation	Outdoor air exposure	Particulate and gaseous contamination; humidity control
Direct-to-chip liquid with dry coolers	Low water use; higher facility water temperatures; heat reuse potential	Dew point margin; heatwave capacity of dry coolers; ERF

Pattern / variant	ENV characteristics	ENV questions that dominate
Direct-to-chip with chillers	Refrigerant charge; trim cooling	Refrigerant safety and leak control; efficiency under part load
Immersion cooling	Dielectric fluids; very high density	Fluid handling, spills, end of life; acoustic conditions differ
Heat reuse to district network	Exported heat	Metered ERF; dependency of off-taker on data-centre operation
Colocation tenant	Provider metrics	Allocation of shared energy and water; reliance on provider data
Hyperscale campus	Utility-scale demand	Long-term grid and water constraints; permits; regulatory reporting

Metric variants: PUE (ISO/IEC 30134-2) with measurement categories determined by where IT energy is metered; partial PUE for sub-systems; WUE (ISO/IEC 30134-9) for site water, with source-side water reported separately; CUE (ISO/IEC 30134-8); REF (ISO/IEC 30134-3); ERF (ISO/IEC 30134-6). Carbon accounting variants: location-based and market-based Scope 2; annual or hourly matching of energy attribute certificates.

4 Required functions and capability linkage

Required Function	Implemented by	Controls
Maintain the environment within equipment envelopes	Sensors, environmental controls, filtration	MCF-ENV-001, MCF-ENV-002, MCF-ENV-003, MCF-ENV-004
Measure and manage energy performance without eroding resilience	Meters, EPMS, energy management system	MCF-ENV-005, MCF-ENV-006, MCF-ENV-007, MCF-ENV-008
Secure, measure and discharge water within limits	Water supply, treatment, meters, interceptors	MCF-ENV-009, MCF-ENV-010, MCF-ENV-011, MCF-ENV-012
Account for carbon and resources traceably	Inventory methodology, registries, reporting systems	MCF-ENV-013, MCF-ENV-014, MCF-ENV-015, MCF-ENV-016
Control hazardous materials and waste	Inventories, storage, detection, contractors	MCF-ENV-017, MCF-ENV-018, MCF-ENV-019, MCF-ENV-020
Withstand climate extremes and resource limits over life	Adaptation plans, procedures, registers	MCF-ENV-021, MCF-ENV-022, MCF-ENV-023, MCF-ENV-024

Capability linkage: environmental conditions link directly to performance Service Outcomes (throttling) and to integrity outcomes (corrosion, condensation, acoustic disruption of storage). Resource continuity links to availability outcomes during heat and drought. Performance claims link to stakeholder obligations captured in GOV and CAP.

DG-ENV-REQ-001 The assessor shall identify, for each ENV control in scope, whether it is assessed as a dependency of a Service Outcome or as support for a stakeholder performance claim, and shall record that linkage in the ControlAssessment.

5 Dependencies, fault domains and common-mode concerns

- **Weather as a common-mode initiator**: a heatwave simultaneously reduces heat-rejection capacity, derates generators and transformers, raises water demand, increases grid stress and may coincide with water restrictions (AIDCF-004 X10). These must be assessed together, not as independent events (MCF-ENV-021, MCF-ENV-011).

- **Single water source**: mains supply from one network, or one borehole, for all evaporative plant.
- **Shared sensing**: one weather station or one outdoor air sensor driving economiser decisions for all air handlers; a common sensor fault becomes a common environmental excursion.
- **Pollution events**: wildfire smoke or industrial releases affect all outdoor air intakes at once (MCF-ENV-003).
- **Metering dependencies**: performance claims depend on a small number of revenue and sub-meters; a single meter fault can distort reported PUE for a year.
- **Regulatory dependencies**: permits for water abstraction, discharge, generator emissions and refrigerant handling are external dependencies whose lapse can constrain operation.
- **Heat reuse off-taker**: where heat is exported, the off-taker depends on the data centre and the data centre may depend on the off-taker's heat sink; this bidirectional dependency must be explicit.

6 Threats, hazards and failure scenarios with propagation paths

Scenario	Propagation path	Primary safeguards (controls)
Extreme ambient plus water restriction (X10)	Heatwave above design ambient → dry coolers lose capacity, evaporative demand rises → water authority restricts supply → heat rejection insufficient → accelerators throttle → training throughput outcome missed	MCF-ENV-021, MCF-ENV-011, MCF-ENV-022
Ambient exceedance at rack inlet	Containment breach or hot-air recirculation → inlet temperatures above envelope → throttling and hardware stress; masked if sensing is at return air	MCF-ENV-001
Condensation	Facility water temperature lowered for capacity → coolant below room dew point in humid season → condensation on manifolds → water on electrical components	MCF-ENV-002
Air contamination	Wildfire smoke drawn in by economisers → corrosive deposition → elevated hardware failure rate over months	MCF-ENV-003
Suppression acoustic event	Inert-gas discharge in a hall with HDD-based storage → acoustic disruption → storage errors and possible data integrity events	MCF-ENV-004
Refrigerant loss	Leak in machinery room → concentration above practical limit → asphyxiation or (A2L) ignition hazard; loss of chiller capacity	MCF-ENV-018
Efficiency-resilience conflict	UPS placed in eco-mode → grid transient during AI load swing → transfer fails → load lost	MCF-ENV-008
Poor coolant or water quality	Biofouling in cooling tower → capacity loss and Legionella risk	MCF-ENV-009
Metering or claim error	PUE computed from an instantaneous snapshot or wrong meters → regulatory report misstated → restatement and loss of trust	MCF-ENV-005, MCF-ENV-016
Battery spares incident	Lithium-ion spares stored at high state of charge next to critical plant → thermal runaway	MCF-ENV-017
Fuel spill	Delivery overfill → fuel to surface drains → environmental incident; fire risk	MCF-ENV-019, MCF-ENV-012
Climate-driven flood (X20)	Rainfall intensity above original design → drainage overwhelmed → flooding of plant	MCF-ENV-023
Resource constraint	Grid reinforcement delayed → planned AI expansion cannot be energised → capability growth outcome missed	MCF-ENV-024

7 Safeguards

- **Prevent**: declared envelopes traceable to ASHRAE classes or OEM limits; dew point margin; filtration and economiser lockouts; quiet suppression nozzles; resilience impact assessment for efficiency measures; water treatment; secondary containment; refrigerant leak checks; safe state of charge for battery spares.
- **Detect**: rack-inlet and coolant-supply sensing; corrosion coupons; refrigerant detectors; water and energy metering; reconciliation of reported metrics.
- **Contain**: interceptors and bunds; quarantine containers for damaged cells; machinery-room ventilation.
- **Correct**: forecast-triggered heatwave procedures; water-restriction procedures; restatement of reported metrics.
- **Recover**: alternative water sources; load-reduction steps; demountable flood defences.

8 Critical-gate linkage

Gate	Name	CRITICAL controls (this domain)	SUPPORTING controls (this domain)
CG-01	Capability and outcome definition	none	MCF-ENV-024
CG-03	Critical dependency visibility	none	MCF-ENV-011, MCF-ENV-022, MCF-ENV-024
CG-04	Life safety and hazardous energy	MCF-ENV-018	MCF-ENV-009, MCF-ENV-017, MCF-ENV-019
CG-05	Resilient electrical supply	none	MCF-ENV-008, MCF-ENV-021
CG-06	Thermal continuity	none	MCF-ENV-001, MCF-ENV-002, MCF-ENV-008, MCF-ENV-009, MCF-ENV-011, MCF-ENV-021
CG-08	Compute, storage and network functional integrity	none	MCF-ENV-001, MCF-ENV-003, MCF-ENV-004
CG-09	Failure containment	none	MCF-ENV-023
CG-14	Change control	none	MCF-ENV-008

Notes on the mapping:

- **CG-04** is carried in ENV by refrigerants and coolants (MCF-ENV-018, CRITICAL), because refrigerant release in occupied or machinery spaces is a direct life-safety hazard. Battery materials, fuel and suppressant handling and water hygiene are SUPPORTING; their in-service safety equivalents are in POW and PHY.
- **CG-06** is informed by six ENV controls (temperature and humidity envelopes, water quality, water scarcity response, heatwave readiness, and the resilience-efficiency trade-off). None is CRITICAL because CG-06 is carried by the THM heat-removal controls; ENV supplies the environmental conditions and stress cases.
- **CG-05** is informed by the resilience-efficiency trade-off (eco-mode, staging) and heatwave readiness (generator derating).
- **CG-08** is informed by environmental conditions that degrade hardware: temperature (throttling), contamination and acoustic effects.
- **CG-01** and **CG-03** are informed by long-term resource constraints and water dependencies.
- Energy, water and carbon performance controls (MCF-ENV-005, MCF-ENV-006, MCF-ENV-007, MCF-ENV-010, MCF-ENV-012 to MCF-ENV-016, MCF-ENV-020) map to no gate. They are scored in the ENV

domain and are decisive for stakeholder performance claims, but not for the capability gates.

DG-ENV-REQ-002 The assessor shall record the gate-level evaluation required for every SUPPORTING ENV control that is NONCONFORMING, stating whether the deficiency defeats the gate scope, before a gate PASS is recorded.

DG-ENV-REQ-003 An ENV performance claim (energy, water, carbon) shall not be reported in an assurance statement unless the corresponding control is CONFORMING at its required claim type, or the claim is reported with the limitation and UNKNOWN Record that apply.

9 Capability-profile parameters

Unlabelled values are attributed to the cited source family; values labelled illustrative show order of magnitude only.

Control	Parameter	Unit	Illustrative default	Basis
MCF-ENV-001	Recommended IT inlet air temperature envelope	°C	18 to 27	ASHRAE TC 9.9 thermal guidelines recommended envelope
MCF-ENV-001	Liquid-cooling facility/TCS supply class	ASHRAE W class	W32 or W45 (illustrative)	ASHRAE TC 9.9 liquid cooling classes; OEM requirement
MCF-ENV-001	Rack inlet sensor density	sensors per rack	3 (top, middle, bottom; illustrative)	Capability profile; ASHRAE measurement practice
MCF-ENV-001	Permitted excursion time outside recommended envelope	hours per year	per OEM allowable (illustrative: 50)	Capability profile
MCF-ENV-002	Recommended dew point envelope	°C dew point	-9 to 15, and ≤ 60% RH	ASHRAE TC 9.9 thermal guidelines recommended envelope
MCF-ENV-002	Minimum margin of coolant supply temperature above room dew point	K	2 (illustrative)	Capability profile; CDU OEM guidance
MCF-ENV-003	Gaseous corrosion severity level	ANSI/ISA-71.04 class	G1	ASHRAE TC 9.9 gaseous and particulate contamination guidance
MCF-ENV-003	Particulate cleanliness	ISO 14644-1 class	8	ASHRAE TC 9.9 contamination guidance
MCF-ENV-003	Corrosion coupon interval	months	3 (illustrative)	Capability profile
MCF-ENV-004	Occupational noise lower / upper exposure action values	dB(A) daily exposure	80 / 85	EU Directive 2003/10/EC (noise at work) or national equivalent
MCF-ENV-004	Sound pressure limit at storage arrays during discharge	dB	per storage OEM threshold	Storage OEM guidance; suppression designer
MCF-ENV-004	Site boundary noise limit	dB(A)	per planning or environmental permit	Permit conditions
MCF-ENV-005	PUE measurement period	months	12	ISO/IEC 30134-2
MCF-ENV-005	PUE measurement category	category	Category 2 or higher (illustrative)	ISO/IEC 30134-2; capability profile

Control	Parameter	Unit	Illustrative default	Basis
MCF-ENV-005	Revenue/sub-meter accuracy class	class	0.5S or better (illustrative)	IEC 62053 series
MCF-ENV-006	Maximum unallocated energy	% of site total	2 (illustrative)	Capability profile
MCF-ENV-007	IT load points for efficiency characterisation	% of design IT load	25 / 50 / 75 / 100 (illustrative)	Capability profile
MCF-ENV-007	Energy reuse factor reporting	ratio	measured annually	ISO/IEC 30134-6
MCF-ENV-009	Legionella risk assessment review interval	months	24 (illustrative)	National Legionella guidance (e.g. HSE L8, ASHRAE 188); capability profile
MCF-ENV-010	WUE measurement period	months	12	ISO/IEC 30134-9
MCF-ENV-011	On-site water storage autonomy at design-day consumption	hours	48 (illustrative)	Capability profile
MCF-ENV-011	Heat-rejection capacity in reduced-water mode at design ambient	% of design IT load	per thermal design (illustrative: 80)	Thermal design basis
MCF-ENV-012	Discharge monitoring frequency	per consent	per permit conditions	Discharge consent
MCF-ENV-013	Materiality threshold for boundary exclusions	% of total emissions	5 (illustrative)	Capability profile; GHG Protocol guidance on exclusions
MCF-ENV-014	Temporal matching granularity	interval	annual or hourly (declared)	GHG Protocol Scope 2 Guidance; hourly matching frameworks
MCF-ENV-015	Embodied data coverage	% of hardware and plant by spend	80 (illustrative)	Capability profile
MCF-ENV-016	Restatement threshold	% change in reported metric	5 (illustrative)	Capability profile
MCF-ENV-017	Maximum state of charge for stored lithium-ion spares	% SoC	30 (illustrative)	OEM guidance; aligns with air-transport limit for UN 3480 lithium-ion batteries
MCF-ENV-018	Refrigerant leak-check frequency	per system	tiered by CO2e charge per applicable regulation	EU F-gas Regulation (EU) 2024/573 or national equivalent
MCF-ENV-018	Refrigerant detector alarm setpoint	% of practical limit or ATEL/ODL	per EN 378 / ISO 5149 (illustrative: 50% of practical limit)	EN 378 / ISO 5149 refrigerating systems safety
MCF-ENV-019	Secondary containment capacity	% of largest tank volume	110 (illustrative)	Common oil-storage regulation and bunding practice; local permit
MCF-ENV-021	Design extreme ambient dry-bulb	°C	site n=20-year extreme (illustrative)	ASHRAE Handbook climatic design data
MCF-ENV-021	Climate-adjusted design ambient uplift	K	2 (illustrative)	Owner-adopted climate projection (e.g. IPCC AR6 scenario)

Control	Parameter	Unit	Illustrative default	Basis
MCF-ENV-021	Forecast trigger margin below design ambient	K	3 (illustrative)	Capability profile
MCF-ENV-022	Water-stress planning horizon	years	20 (illustrative)	Capability profile
MCF-ENV-023	Climate scenario and horizon for flood adaptation	scenario / year	SSP2-4.5 to 2060 (illustrative)	IPCC AR6 scenarios; owner-adopted projection
MCF-ENV-023	Demountable defence deployment time	hours	less than flood warning lead time (illustrative: 4)	Flood response plan
MCF-ENV-024	Resource planning horizon	years	15 (illustrative)	Capability profile

10 Evidence expectations by claim type

Claim type	Assurance Level	Controls	Profile tiers used	Sufficiency floors used
DESIGN	AL2	MCF-ENV-008, MCF-ENV-013, MCF-ENV-015, MCF-ENV-022, MCF-ENV-023, MCF-ENV-024	CP-1, CP-2	S2, S3
IMPLEMENTATION	AL3	MCF-ENV-004, MCF-ENV-006, MCF-ENV-014	CP-2	S3
OPERATION	AL4	MCF-ENV-001, MCF-ENV-002, MCF-ENV-003, MCF-ENV-005, MCF-ENV-007, MCF-ENV-009, MCF-ENV-010, MCF-ENV-012, MCF-ENV-016, MCF-ENV-017, MCF-ENV-018, MCF-ENV-019, MCF-ENV-020	CP-3, CP-5	S3, S4
RESILIENCE	AL5	MCF-ENV-011, MCF-ENV-021	CP-4	S4

- **DESIGN (AL2)**: methodology and boundary statements, LCA methods, climate and water-stress assessments, resilience impact assessments (E1 with E7 analysis). Climate assessments name the scenario, horizon and data source.
- **IMPLEMENTATION (AL3)**: meter-to-category mapping, allocation methods reconciled to meters, registry retirement statements, acoustic and noise surveys as built (E1+E2 with reconciliation).
- **OPERATION (AL4)**: twelve months of telemetry and meter data, coupon results, leak-check records, treatment records, inventories reconciled to physical counts (E3, often with E2 configuration evidence).
- **RESILIENCE (AL5)**: exercises of water-restriction switchover and heatwave procedures, or records of real events at or beyond design conditions (E4).

Evidence cautions:

- PUE and WUE are annual energy and water ratios; instantaneous or monthly snapshots do not satisfy ISO/IEC 30134-2 or -9.
- Environmental telemetry is E3 only if sensor location and calibration are established; return-air sensing does not evidence inlet temperature (U8).
- Certificates of renewable energy are evidence only when their retirement is verified in the registry (U6 otherwise).
- Supplier sustainability reports are E5 at best for their own scope; they rarely evidence the site's metrics.

DG-ENV-REQ-004 Reported energy and water metrics relied on in an assurance statement shall be reperformed by the assessor from source meter data for at least the reporting period, and the reperformance shall be retained as E7 evidence with lineage to the meters.

11 Assessment guidance

1. **Separate the two faces.** Decide for each control whether it supports outcomes or performance claims (DG-ENV-REQ-001).
2. **Check envelopes at the equipment.** Confirm sensor locations at rack inlet and TCS supply; compute excursion time and, for liquid-cooled halls, dew point margin.
3. **Stress the climate.** Take the design extreme ambient and the climate-adjusted ambient; check heat rejection, generators and transformers together, then add water restriction.
4. **Find the efficiency measures.** Read UPS modes, setpoints and staging from configuration, not from policy; check each has a resilience impact assessment.
5. **Reperform the metrics.** Recompute PUE, WUE and CUE from meter data; reconcile allocation to the site total; verify certificate retirement.
6. **Inspect hazardous materials.** Reconcile refrigerant, coolant and battery inventories to nameplates and stores; check detection, ventilation, containment and quarantine.
7. **Read the permits.** List environmental permits with conditions and renewal dates; compare monitoring data with limits.
8. **Roll up.** Record results, Findings and UNKNOWN Records; evaluate CG-04 for MCF-ENV-018 and supply gate-level evaluations for SUPPORTING mappings.

Common pitfalls: accepting design PUE as measured; omitting generator fuel from total facility energy; counting refrigerant leakage outside Scope 1; ignoring A2L requirements when refrigerants are changed; treating dry-cooler capacity at ISO conditions as available in a heatwave.

12 Testing safety (hazardous-energy testing authorisation)

Most ENV assessment is desk-based or observational. The exceptions involve hazardous atmospheres, chemicals and plant operation.

DG-ENV-REQ-005 An assessor shall not take water, fuel or refrigerant samples, operate plant to change environmental conditions, or enter machinery rooms during a refrigerant alarm; sampling and tests relied on as evidence shall have been performed by competent persons under the site's procedures.

DG-ENV-REQ-006 Exercises that change cooling mode (for example switchover to reduced-water or dry operation) relied on as evidence shall have been performed under a method statement approved by the facility operator, with thermal contingency in place and the capability owner informed.

Activity	Hazard	Precondition
Refrigerant detector test (MCF-ENV-018)	Calibration gas; alarm-triggered ventilation	Certified contractor; machinery room cleared
Legionella sampling (MCF-ENV-009)	Aerosol exposure	Respiratory protection per control scheme
Fuel sampling (MCF-ENV-019 with POW)	Flammable liquid	Ignition-source control; spill kit
Reduced-water switchover (MCF-ENV-011)	Thermal margin loss	Method statement; thermal contingency
Occupational noise survey (MCF-ENV-004)	Noise exposure	Hearing protection
Interceptor inspection (MCF-ENV-012)	Confined space	Inspection from surface only

13 UNKNOWN hot-spots

Control	UNKNOWN rule
MCF-ENV-001 Temperature Control	If inlet sensing is only at return air, record a U8 method UNKNOWN; the OPERATION claim is UNKNOWN for that hall.
MCF-ENV-002 Humidity Control	If dew point margin is not monitored for a liquid-cooled hall, record a U1 UNKNOWN and refer to THM fluid controls.
MCF-ENV-005 PUE Measurement	If metering does not support the declared category, record a U8 method UNKNOWN and report PUE only with a limitation.
MCF-ENV-008 Resilience-Efficiency Trade-Off	If an efficiency measure is found in configuration without an assessment, record a U1 UNKNOWN and carry it into CG-05/CG-06 gate-level evaluation.
MCF-ENV-011 Water Scarcity Response	If reduced-water capacity is unknown, record a U1 UNKNOWN referred to CG-06.
MCF-ENV-014 Energy Attribute Evidence	If retirement cannot be verified in the registry, record a U6 source-unverified UNKNOWN on the claim.
MCF-ENV-018 Refrigerants and Coolants	If detection in a machinery room is untested or absent where required, record a US4 UNKNOWN against CG-04.
MCF-ENV-021 Heatwave Readiness	If capacity at design ambient is unevaluated, record a U1 UNKNOWN referred to CG-06 and CG-05.

Typical causes: colocation providers supplying summary metrics without meter data (U2); sensors at return air rather than inlet (U8); no evaluation of capacity at design ambient (U1); efficiency measures configured without assessment (U1); certificates bought but retirement not visible in the registry (U6).

DG-ENV-REQ-007 Any UNKNOWN affecting MCF-ENV-018 shall be recorded with severity US4 and shall set CG-04 to UNKNOWN until dispositioned; UNKNOWNs on ENV performance claims shall be disclosed with the claim.

14 Interfaces with other domains

Domain	Interface	What ENV needs	What ENV provides
THM	Cooling	Plant capacity curves; setpoints; refrigerant inventory	Envelopes; dew point margin; climate stress cases; water constraints
POW	Energy and batteries	Meter data; UPS mode; generator fuel use; battery inventory	Resilience impact of efficiency measures; heatwave derating case
PHY	Site and fire	Site hazard assessment; suppression design	Climate-adjusted flood; acoustic constraints on suppression
CMP	Hardware	Throttling telemetry; OEM envelopes	Environmental excursion data
STO	Storage	Acoustic and vibration sensitivity	Discharge acoustic design
CAP	Outcomes and growth	Demand forecasts	Resource constraints on growth
GOV	Reporting	Stakeholder and regulatory obligations	Traceable metrics
SUP	Recyclers, suppliers	Authorised waste operators; OEM product footprints	Disposition records
EVA	Evidence	Provenance rules	Reperformed metrics with lineage

15 Informative references

Document-level references; no clause-level reliance is implied.

- ISO/IEC 30134 series — Data centre key performance indicators: -2 PUE, -3 REF, -6 ERF, -8 CUE, -9 WUE (EN 50600-4-x equivalents)
- ASHRAE TC 9.9 — Thermal Guidelines for Data Processing Environments; liquid cooling guidelines; gaseous and particulate contamination guidance
- ANSI/ISA-71.04 — Environmental conditions for process measurement and control systems: airborne contaminants
- ISO 14644-1 — Cleanroom air cleanliness classification
- EN 50600-2-3 / ISO/IEC 22237 — Environmental control
- ISO 14001 — Environmental management systems; ISO 50001 — Energy management systems
- GHG Protocol Corporate Standard and Scope 2 Guidance; ISO 14064-1 and -3
- ISO 14040 / ISO 14044 — Life cycle assessment; EN 15978; ISO 14067
- ISO 14046 — Water footprint; WRI Aqueduct water-risk atlas
- ISO 14090 / ISO 14091 — Adaptation to climate change; IPCC AR6
- ASHRAE Handbook — Fundamentals (climatic design conditions)
- EN 378 / ISO 5149 — Refrigerating systems safety; ISO 817 / ASHRAE 34 — refrigerant safety classification; ASHRAE 15
- EU F-gas Regulation (EU) 2024/573; EU Battery Regulation (EU) 2023/1542; EU WEEE Directive 2012/19/EU
- EU Energy Efficiency Directive (EU) 2023/1791 and Delegated Regulation (EU) 2024/1364 on the data-centre rating scheme
- EU Directive 2003/10/EC — noise at work; ISO 9613-2; ISO 7779
- ASHRAE 188 and HSE L8 — Legionella risk management
- EU Code of Conduct on Data Centre Energy Efficiency — best practice guidelines

16 Control summary

Control ID	Title	Control type	Required AL	Claim type	Gate mappings
MCF-ENV-001	Temperature Control	DETECTIVE	AL4	OPERATION	CG-06 SUPPORTING; CG-08 SUPPORTING
MCF-ENV-002	Humidity Control	PREVENTIVE	AL4	OPERATION	CG-06 SUPPORTING
MCF-ENV-003	Particulate and Corrosion	PREVENTIVE	AL4	OPERATION	CG-08 SUPPORTING
MCF-ENV-004	Vibration and Acoustic Conditions	PREVENTIVE	AL3	IMPLEMENTATION	CG-08 SUPPORTING
MCF-ENV-005	PUE Measurement	DETECTIVE	AL4	OPERATION	none
MCF-ENV-006	Energy Allocation	DIRECTIVE	AL3	IMPLEMENTATION	none
MCF-ENV-007	Efficiency Under Load	DETECTIVE	AL4	OPERATION	none
MCF-ENV-008	Resilience-Efficiency Trade-Off	DIRECTIVE	AL2	DESIGN	CG-05 SUPPORTING; CG-06 SUPPORTING; CG-14 SUPPORTING

Control ID	Title	Control type	Required AL	Claim type	Gate mappings
MCF-ENV-009	Water Source and Quality	PREVENTIVE	AL4	OPERATION	CG-06 SUPPORTING; CG-04 SUPPORTING
MCF-ENV-010	WUE Measurement	DETECTIVE	AL4	OPERATION	none
MCF-ENV-011	Water Scarcity Response	RECOVERY	AL5	RESILIENCE	CG-06 SUPPORTING; CG-03 SUPPORTING
MCF-ENV-012	Discharge Control	PREVENTIVE	AL4	OPERATION	none
MCF-ENV-013	Carbon Accounting Boundary	DIRECTIVE	AL2	DESIGN	none
MCF-ENV-014	Energy Attribute Evidence	DETECTIVE	AL3	IMPLEMENTATION	none
MCF-ENV-015	Embodied Resource Data	DIRECTIVE	AL2	DESIGN	none
MCF-ENV-016	Reporting Traceability	DETECTIVE	AL4	OPERATION	none
MCF-ENV-017	Battery Materials	PREVENTIVE	AL4	OPERATION	CG-04 SUPPORTING
MCF-ENV-018	Refrigerants and Coolants	PREVENTIVE	AL4	OPERATION	CG-04 CRITICAL
MCF-ENV-019	Fuel and Suppressant Handling	PREVENTIVE	AL4	OPERATION	CG-04 SUPPORTING
MCF-ENV-020	Hardware and Media Waste	DIRECTIVE	AL4	OPERATION	none
MCF-ENV-021	Heatwave Readiness	RECOVERY	AL5	RESILIENCE	CG-06 SUPPORTING; CG-05 SUPPORTING
MCF-ENV-022	Drought Readiness	DIRECTIVE	AL2	DESIGN	CG-03 SUPPORTING
MCF-ENV-023	Flood Adaptation	PREVENTIVE	AL2	DESIGN	CG-09 SUPPORTING
MCF-ENV-024	Long-Term Resource Constraints	DIRECTIVE	AL2	DESIGN	CG-01 SUPPORTING; CG-03 SUPPORTING

17 Worked synthetic example

Synthetic example. The site, figures and results below are invented for illustration and do not describe any real facility or assessment.

Context. Synthetic Site S-3 is a direct-to-chip liquid-cooled site with adiabatic-assisted dry coolers, two chillers for trim cooling using an A2L refrigerant, and a published annual PUE. The decision question has two parts: (a) can S-3 sustain the training Service Outcome through the declared heatwave and water-restriction scenario; (b) is the published PUE fit to be cited in the operator's regulatory report. Profile parameters (illustrative for the example): design extreme ambient taken from the site's n=20-year value; climate uplift 2 K; water storage autonomy 48 hours; dew point margin 2 K.

Selected results.

Control	Claim type	Synthetic observation	Result
MCF-ENV-001	OPERATION	Rack-inlet sensors in place; excursion time within limit; coolant supply sensors at CDU secondary outlet	CONFORMING
MCF-ENV-002	OPERATION	Dew point margin fell to 0.4 K for 30 hours in August after facility water setpoint was lowered	NONCONFORMING
MCF-ENV-005	OPERATION	Reperformed PUE from meters differs by 0.03 from published value; generator fuel energy omitted	PARTIAL
MCF-ENV-008	DESIGN	Lowered facility water setpoint (efficiency/capacity change) made without resilience impact assessment	NONCONFORMING
MCF-ENV-011	RESILIENCE	Storage autonomy recomputed at 31 hours at design-day consumption; dry-mode capacity 78% of IT load; exercise performed	PARTIAL
MCF-ENV-018	OPERATION	A2L chiller room has detection and ventilation tested; leak checks current	CONFORMING
MCF-ENV-021	RESILIENCE	Capacity analysis at climate-adjusted ambient not performed	UNKNOWN (U1)

Gate-level evaluations. CG-04: MCF-ENV-018 CONFORMING; ENV contributes no defeat. CG-06 (SUPPORTING inputs): MCF-ENV-002 NONCONFORMING and MCF-ENV-008 NONCONFORMING are referred to the THM-led CG-06 evaluation, which must decide whether condensation risk and the unassessed setpoint change defeat thermal continuity; MCF-ENV-011 PARTIAL and the MCF-ENV-021 UNKNOWN mean that the heatwave-plus-restriction scenario is not shown to be sustained, and the CG-06 evaluation records this as a material condition. CG-14: MCF-ENV-008 NONCONFORMING indicates an uncontrolled material change, evaluated at CG-14 gate level.

Performance claim. Under DG-ENV-REQ-003 the published PUE is not cited without a limitation: the reperformed value including generator fuel energy is reported, with the PARTIAL result disclosed.

Actions. Restore dew point margin or add room dehumidification; perform a resilience impact assessment of the setpoint change under change control; increase water storage or reduce design-day consumption; complete capacity analysis at climate-adjusted ambient; correct the PUE calculation.